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Physics HL

Internal Assessment

Investigating the Effect of Bob Radius on Damping Constant

in a Simple Pendulum

Word Count: 3000

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1 Introduction

1.1 Introduction

A pendulum is a device that consists of a bob suspended from a fixed point so that it oscillates freely, and it has a wide range of real-life applications from mechanical clocks to dumpers in skyscrapers. Although energy loss due to factors such as drag and friction is often neglected in theories, its effect is non-negligible in real-life, as it causes the amplitude to decay over time. To mathematically express such phenomena, damping constant, often denoted by letter b, is commonly used as the measure of the rate at which amplitude decays. Thus, this investigation will focus on one of the factors causing such damping effect, that is, drag force on the bob, to explore the relationship between bob radius and damping constant.

1.2 Research Question

How does varying the bob radius (7.0, 14.0, 19.0, 24.0, 29.0mm) of a simple pendulum affect the damping constant, calculated by photogate-time measurements taken after 49.75 oscillations?

1.3 Background Information

1.3.1 The Nature of an Ideal Simple Pendulum

In physics, a simple pendulum is a system consisting of a bob suspended to a fixed point by a non-extensible string, where the bob moves on a vertical two-dimensional plane**Invalid source specified.**.

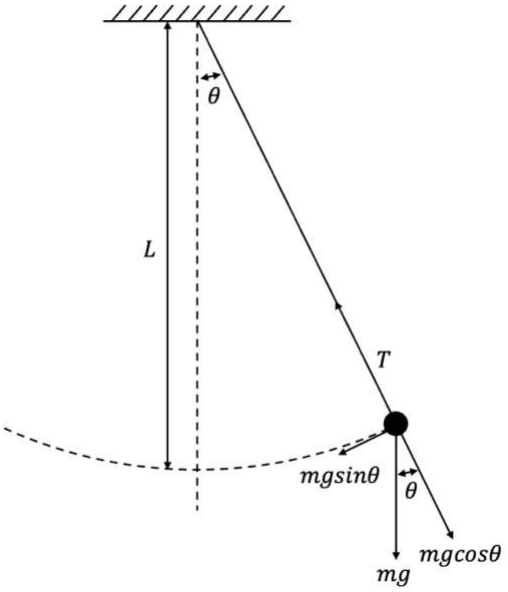


Figure 1: Free Body Diagram of an Ideal Simple Pendulum

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Without damping effect or external driving forces, the motion of a simple pendulum solely relies on the weight mg acting on the bob, where m is the mass of the bob and g is the gravitational acceleration. Weight can be further resolved into vertical and horizontal components in terms of the angular displacement θ from the equilibrium position.

Since the tension force T and the vertical component of weight $mgcos\theta$ balances each other, the restoring force F_r acting on the bob is

$$F_r = -mgsin\theta \tag{1}$$

For a small angular displacement, the linear displacement x of the bob from its equilibrium position can be approximated by the horizontal displacement**Invalid source specified.**, which can be written in terms of string length l and angular displacement θ as

$$x = lsin\theta \tag{2}$$

Rearranging Eq. (2) and substituting it into Eq. (1) yields

$$F_r = -\frac{mg}{l}x \tag{3}$$

This Eq. (3) implies that, an ideal simple pendulum without damping effect or external driving forces demonstrates simple harmonic motion characterized by $F_r \propto -x$.

1.3.2 Damping Effect in a Simple Pendulum

In the case of a small-sized simple pendulum, since the bob is spherical and travels at a low speed, the drag force F_{drag} acting on it can be approximated by the Stoke's law**Invalid source specified.**

$$F_{drag} = -6\pi\eta rv \tag{4}$$

For convenience, the term $6\pi\eta r$ is denoted by the constant γ

$$F_{drag} = -\gamma v \tag{5}$$

Using Eq. (3) and (5), the net force F_{damped} acting on the bob in a damped simple pendulum is given by

$$F_{damped} = F_r + F_{drag} \tag{6}$$

$$ma = -\frac{mg}{l}x - \gamma v \tag{7}$$

where a is the acceleration of bob (ms^{-2}).

Eq. (7) can be rearranged into a differential form with respect to time t as

$$ma + \gamma v + \frac{mg}{l}x = 0 \tag{8}$$

$$m\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + \frac{mg}{l}x = 0 \tag{9}$$

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By considering the case of light damping**Invalid source specified.**, as shown in Appendix 6.1, the solution to this differential equation is given by

$$x(t) = x_0 e^{-\frac{\gamma}{2m}t} \cos(\omega' t - \phi) = x_0 e^{-bt} \cos(\omega' t - \phi) \tag{10}$$

where x_0 is initial amplitude (m), ω' is damped angular frequency (rad s^{-1}), ϕ is phase difference (rad), t is time (s), and b is damping constant (kg $m^{-1}s^{-1}$) given by $b = \frac{\gamma}{2m}$.

This Eq. (10) describes the motion of a damped simple pendulum. Since displacement x has a local maximum at $t = 0$, this Eq. (10) assumes that the bob is released at $x = x_0$ at $t = 0$. Thus, time t specifically refers to the time after bob is released at $x = x_0$.

The term $\frac{\gamma}{2m}$ in Eq. (10) is commonly denoted as a damping constant b. Therefore, for a simple pendulum with a spherical bob, substituting $\gamma = 6\pi r\eta$ gives an expression for damping constant as

$$b = \frac{\gamma}{2m} = \frac{6\pi r\eta}{2m} = \frac{3\pi r\eta}{m} \tag{11}$$

1.3.3 Measuring Damping Constant with Photogate

The Eq. (10) suggests that the bob will have its maximum displacement (amplitude) x_{max} at angular displacement of $\omega' t = k\pi$ where $k \in \mathbb{Z}$.

Since $\cos(k\pi) = 1$, amplitude of the system at given time t is expressed as

$$x_{max} = x_0 e^{-bt} \tag{12}$$

While Eq. (12) shows the relationship between damping constant b, initial displacement x_0, and amplitude x_{max} at a given time t, it is difficult to directly measure amplitude while a simple pendulum is in motion. Therefore, damping constant needs to be expressed in terms of other variables that are readily measurable in the experiment.

As shown in Appendix 6.2, Eq. (12) can be used to derive an expression for b as

$$b = -\frac{\ln\left(\frac{2r}{\Delta t\sqrt{lg}sin\theta_0}\right)}{TN} \tag{13}$$

where r is bob radius (m), Δt is the time taken for the bob to move past the equilibrium position (s), l is string length (m), θ_0 is initial angular displacement, T is the period of pendulum, and N is the number of completed oscillations. Since g is a constant, with this Eq. (13), damping constant b can be calculated by measuring Δt for given values of r, l, θ_0, T, N.

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1.4 Hypothesis

It can be hypothesized that, as the bob radius increases, damping constant also increases proportionally for a simple pendulum.

This is because the drag force acting on the bob should follow the Stoke's law. The drag force should increase proportionally with increasing bob radius, which results in a greater damping effect.

This hypothesis is also supported by the theoretical relationship between damping constant and bob radius in Eq. (11), which implies the relationship $b \propto r$:

$$b = \frac{3\pi r\eta}{m} \tag{14}$$

However, it should be noted that this hypothesis is conditional on the Stoke's law being valid under the experimental conditions.

2 Methodology

2.1 Variables

2.1.1 Independent and Dependent Variable

Independent variable: Bob radius (r) (7.0, 14.0, 19.0, 24.0, 29.0mm)

This variable will be manipulated by covering a brass bob of radius 7mm with two Styrofoam hemispheres of different radius (14.0, 19.0, 24.0, 29.0mm). All bob radii were measured with a vernier caliper to the uncertainty of ±0.1mm. These five increments were chosen based on the availability of Styrofoam hemispheres, and the width of photogate (75.0mm) which sets the upper limit of bob diameter. (Brass bobs weigh 13.80g, while a pair of Styrofoam hemispheres weigh 0.2-0.6g. Thus, percentage change in the mass of bob due to the addition of Styrofoam hemispheres is about 2-5%, which is negligible compared to the random errors)

Dependent variable: Damping constant (b)

This variable will be calculated by the time taken for the bob to move past the equilibrium position Δt after 49.75 completed oscillations (at $N = 49.75$), and period T. This value of N was chosen based on a preliminary trial, which showed that the amplitude dies out approximately after 60 completed oscillations. The value of Δt is measured using a photogate, by recording the duration for which the laser is blocked. Thus, time in gate Δt is given by

$$\Delta t = t_{unblocked} - t_{blocked} \tag{15}$$

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where $t_{blocked}$ and $t_{unblocked}$ are the times at which the laser gate becomes blocked or unblocked (s), respectively. The photogate measures $t_{blocked}$ and $t_{unblocked}$ to the uncertainty of ±0.00001s.

Period T is also calculated using time after release t at which 49.75 oscillations are completed by

$$T = \frac{t}{49.75} \tag{16}$$

where t is calculated as the average of $t_{blocked}$ and $t_{unblocked}$ as

$$t = \frac{t_{blocked} + t_{unblocked}}{2} \tag{17}$$

2.1.2 Controlled Variables

Controlled variable	Reason for control	Method of control
Initial amplitude (Initial angular displacement and string length)	Initial amplitude determines the maximum velocity of the bob. Since drag force depends on velocity, changing initial amplitude would affect damping constant.	Initial amplitude is fixed at $40 \sin\left(\frac{\pi}{18}\right) \approx 7$cm, by keeping initial angular displacement and string length constant at $\frac{\pi}{18}$ rad and 40cm, respectively.
Shape of bob	Shape of bob determines the flow characteristics around it, which affects drag force and thus damping constant.	Spherical bobs are used throughout experiment to ensure valid application of Stoke's law.
Initial velocity and acceleration (Manner of release)	The model used assumes that the bob is released from rest, with acceleration solely due to gravity. Thus, any addition of force would lead to an incorrect calculation of the damping constant.	Bob is initially positioned with a ruler. Once the bob is stationary, the ruler is dropped to release the bob. This minimizes additional force exerted on the bob by hand.
Mass of bob	The damping constant is inversely proportional to mass ($b \propto \frac{1}{m}$) as in Eq. (11). Thus, change in mass would affect damping constant, which makes it difficult to isolate the effect of bob radius.	Brass bobs of the same mass are used. The bob radius was varied with Styrofoam hemispheres of negligible mass to minimize change in mass.

Table 1: List of Controlled Variables

2.1.3 Key Assumptions

- **Damping effect is predominantly due to drag force on bob:**
  The damping effect due to friction between string and pivot, as well as the drag force on string is assumed negligible.
- **Drag force on bob is predominantly due to viscous drag:**
  Since the bob travels at low velocity, the airflow is non-turbulent and therefore the effect of pressure drag is assumed negligible.
- **The small angle approximation:**
  The bob's angular displacement is small enough that the bob's linear displacement can be approximated by its horizontal displacement. To maintain the validity of this assumption, the initial angular displacement is maintained small ($\frac{\pi}{18}$ rad).
- **Constant period:**
  Since the period of a ideal simple pendulum is only dependent on string length and gravitational acceleration, it is assumed that the period of a dumped pendulum in each trial is independent of time after release for analytical purpose. This assumption is confirmed valid in a preliminary trial.
- **2D motion of bob:**
  The bob moves in a 2D plane and any 3D motion is neglected, per the definition of a simple pendulum.

2.2 Apparatus

Equipment	Quantity	Equipment	Quantity
Brass bob (r = 7.0 ±0.1mm, m = 13.80 ±0.01g)	5	Stand	1
Styrofoam hemisphere (r = 14.0, 19.0, 24.0, 29.0 ±0.1mm)	2 for each size	Clamp	1
LabQuest	1	Ring Clamp	2
Photogate (±0.00001s)	1	String	5.0m
Protractor (±$\frac{\pi}{36}$ rad)	1	Double-sided tape	1.0m
60cm Ruler (±0.5mm)	1	Utility Cutter	1
Vernier Caliper (±0.1mm)	1	Electronic scale (±0.01g)	1

Table 2: List of Apparatus

2.3 Procedure

2.3.1 Preliminary Trial

A preliminary trial was conducted prior to the main experiment to determine the appropriate number of completed oscillations N after which $t_{blocked}$ and $t_{unblocked}$ are measured.

To test this, a simple pendulum of string length 40cm was constructed with bobs of different radius (7.0, 14.0, 19.0, 24.0, 29.0mm) to test the number of completed oscillations before the amplitude dies out.

After the preliminary trial, it was found that the amplitude dies out after approximately 60 complete oscillations at fastest. Since the bob keeps blocking the laser gate and hence Δt cannot be measured reliably once amplitude is too small, it was decided that the measurements of $t_{blocked}$ and $t_{unblocked}$ are made after 49.75 complete oscillations.

2.3.2 Data Collection

1. Using a utility cutter, create a hemisphere-shaped hole of radius 7.0mm at the center of Styrofoam hemispheres
2. Using doubles-sided tape, cover a brass bob with two Styrofoam hemispheres to create five bobs of different radius (7.0, 14.0, 19.0, 24.0, 29.0mm) in total. Attach each bob to a 1.0m long string.
3. Construct a simple pendulum following the diagrams below. Make sure that the center of bob is positioned exactly at the same height as the laser gate.
4. Turn on LabQuest. Set the Mode to “Photogate timing”, Photogate Mode to “Gate”, and End of Data Collection to “After 200 events.”
5. Displace the bob to a side by $\frac{\pi}{18}$ rad using a ruler. Wait until the bob comes to rest.
6. Start data collection on the LabQuest and drop the ruler to release the bob.
7. Once the LabQuest automatically stops data collection after 200 events, record the last time at which gate became blocked or unblocked.
8. Repeat step 5-7 to complete 5 trials in total.
9. Repeat step 5-8 with different bob radius (7.0, 14.0, 19.0, 24.0, 29.0mm).

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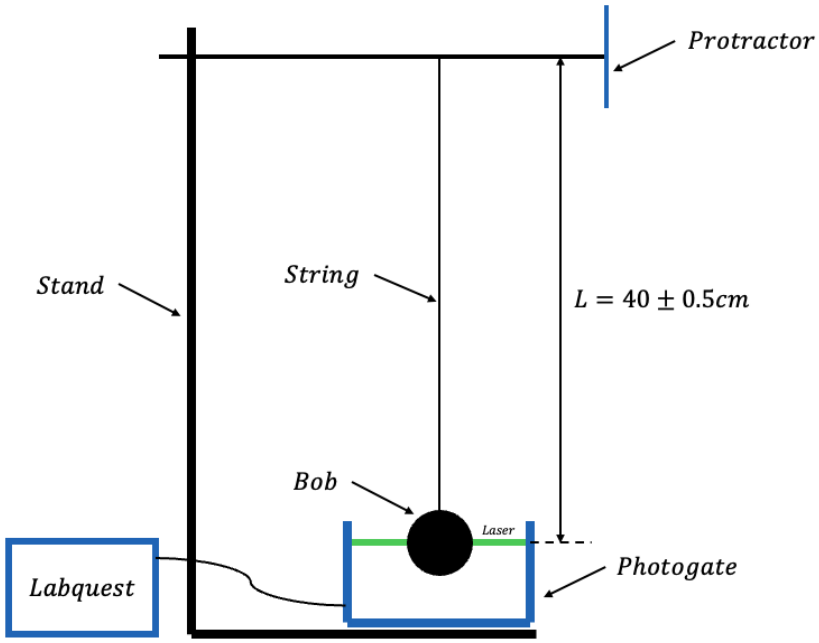


Figure 2: Side view of the Experimental Set-up

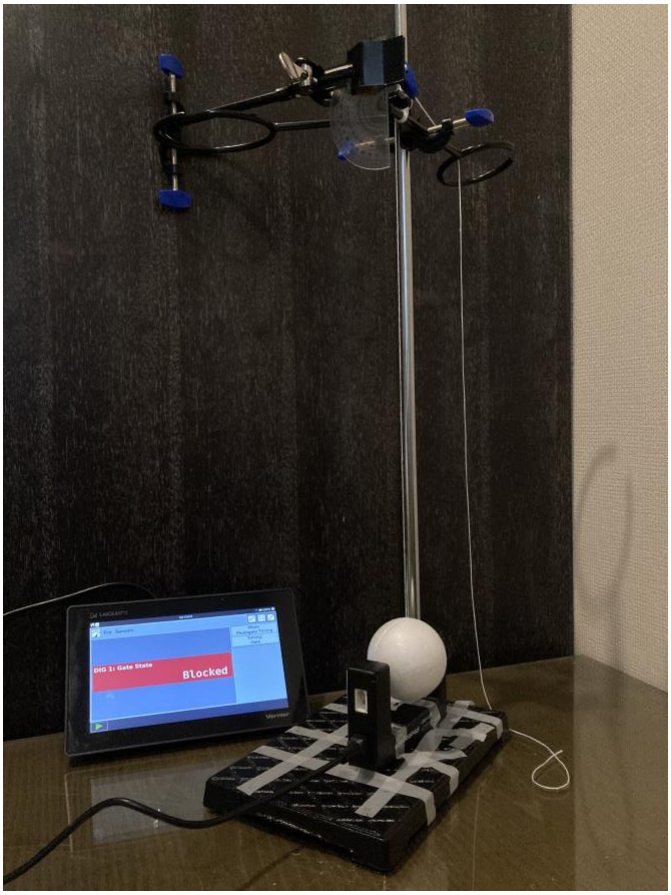


Figure 3: An Example of the Experimental Set-up

2.4 Risk Assessments

Hazards	Measures taken
Safety – Utility knife can cut skins, causing injuries. Fragments can also hit eyes when a blade break**Invalid source specified.**.	Wear protective goggles and gloves. Always use a sharp blade and don't apply excessive pressure on the blade.**Invalid source specified.**.
Safety – LED laser used in photogate can damage eyes.	Do not directly look at the laser source of photogate.
Ethical/Environmental – There are no ethical or environmental concerns, as no animals, human beings, or contaminating chemicals are directly involved in this experiment.	N/A

Table 3: Risk Assessments

3 Data Analysis

3.1 Qualitative Data

For all trials, the damping effect was observed as the amplitude gradually decreased over time until it died completely after approximately 80 seconds. As the radius of the bob increased, the rate of the decay of amplitude evidently increased. In addition, in some of the trials, the bob's movement was not completely on a 2D plane, but rather a 3D motion that resembles an ellipse.

3.2 Quantitative Data

3.2.1 Raw Data Table

Bob Radius r (±0.1 mm)	Trial No.	Time after release (±0.00001s)	
		$t_{blocked}$	$t_{unblocked}$
7.0	1	62.48052	62.52847
	2	62.45600	62.50828
	3	62.47025	62.52166
	4	62.47684	62.52295
	5	62.46548	62.51383
	6	62.48675	62.53263
	7	62.49114	62.53632
	8	62.46382	62.51025
	9	62.44907	62.49968
	10	62.45905	62.50819
14.0	1	62.58804	62.69038
	2	62.58826	62.69523
	3	62.59482	62.69765
	4	62.60400	62.71176
	5	62.62042	62.72192
	6	62.62340	62.72576
	7	62.63243	62.73256
	8	62.60942	62.72077
	9	62.46819	62.57054
	10	62.47736	62.58238
19.0	1	62.55693	62.72248
	2	62.58194	62.74272
	3	62.60636	62.76014
	4	62.58368	62.75181
	5	62.59007	62.75243
	6	62.59821	62.76522
	7	62.61552	62.77576
	8	62.60661	62.77908
	9	62.61182	62.77955
	10	62.61735	62.78171
24.0	1	62.55438	62.78483
	2	62.55232	62.79237
	3	62.58651	62.82115
	4	62.59596	62.82267
	5	62.58612	62.82569
	6	62.60633	62.82996
	7	62.58684	62.82367
	8	62.58843	62.82149
	9	62.60227	62.82801
	10	62.61098	62.83736
29.0	1	62.69777	63.07698
	2	62.73601	63.06345
	3	62.74401	63.05570
	4	62.74401	63.05936
	5	62.72194	63.07639
	6	62.74800	63.06998
	7	62.75361	63.06771
	8	62.75164	63.07067
	9	62.74326	63.07885
	10	62.72070	63.07509

Table 4: Raw Data Table

3.2.2 Processed Data Tables

Bob Radius r (±0.1mm)	Trial No.	Time in gate Δt (±0.00002s)	Period T (±0.00001s)
7.0	1	0.04795	1.256372
	2	0.05228	1.255922
	3	0.05141	1.256200
	4	0.04611	1.256279
	5	0.04835	1.256073
	6	0.04588	1.256476
	7	0.04518	1.256557
	8	0.04643	1.256021
	9	0.05061	1.255766
	10	0.04914	1.255952
14.0	1	0.10234	1.259080
	2	0.10697	1.259131
	3	0.10283	1.259221
	4	0.10776	1.259455
	5	0.10150	1.259722
	6	0.10236	1.259791
	7	0.10013	1.259950
	8	0.11135	1.259600
	9	0.10235	1.256671
	10	0.10502	1.256882
19.0	1	0.16555	1.259090
	2	0.16078	1.259544
	3	0.15378	1.259965
	4	0.16813	1.259653
	5	0.16236	1.259724
	6	0.16701	1.259934
	7	0.16024	1.260214
	8	0.17247	1.260158
	9	0.16773	1.260215
	10	0.16436	1.260292
24.0	1	0.23045	1.259691
	2	0.24005	1.259746
	3	0.23464	1.260378
	4	0.22671	1.260489
	5	0.23957	1.260420
	6	0.22363	1.260666
	7	0.23683	1.260407
	8	0.23306	1.260401
	9	0.22574	1.260606
	10	0.22638	1.260787
29.0	1	0.37921	1.264068
	2	0.32744	1.264316
	3	0.31169	1.264319
	4	0.31535	1.264355
	5	0.35445	1.264305
	6	0.32198	1.264502
	7	0.31410	1.264536
	8	0.31903	1.264546
	9	0.33559	1.264544
	10	0.35439	1.264279

Table 5: Processed Data Table 1

Bob Radius r (±0.1 mm)	Time in gate Δt		Period T	
	Average time in gate $\overline{\Delta t}$ (s)	Standard deviation $\sigma_{\Delta t}$ (s)	Average period $\overline{T}$ (s)	Standard deviation σ_T (s)
7.0	0.04833 ±7%	0.00248	1.25616 ±0.03%	0.00026
14.0	0.10426 ±5%	0.00347	1.25895 ±0.13%	0.00118
19.0	0.16424 ±6%	0.00522	1.25988 ±0.05%	0.00038
24.0	0.23171 ±4%	0.00600	1.26036 ±0.04%	0.00036
29.0	0.33332 ±10%	0.02243	1.26438 ±0.02%	0.00015

Table 6: Processed Data Table 2

Bob Radius r (±0.1 mm)	Damping constant b (kg m^{-1}s^{-1})	Theoretical value (kg m^{-1}s^{-1})	Percentage error (%)
7.0	0.0027 ±8%	0.000088 ±1.5%	3000
14.0	0.0039 ±8%	0.00018 ±0.8%	2100
19.0	0.0063 ±9%	0.00024 ±0.6%	2500
24.0	0.0081 ±10%	0.00030 ±0.5%	2600
29.0	0.0108 ±12%	0.00037 ±0.4%	2900

Table 7: Processed Data Table 3

3.2.3 Formulas, Sample Calculations, Uncertainties

Calculation Type	Formula Sample calculation
	Uncertainty Formula Sample uncertainty calculation
Time in gate, Δt	$\Delta t = t_{unblocked} - t_{blocked}$ $\Delta t = 62.52847 - 62.48052 = 0.04795s$
	$\delta\Delta t = \delta t_{unblocked} + \delta t_{blocked}$ $\delta\Delta t = 0.00001 + 0.00001 = 0.00002s$
Average time in gate, $\overline{\Delta t}$	$\overline{\Delta t} = \frac{\sum \Delta t}{10}$ $\overline{\Delta t} = \frac{0.04795 + 0.05228 + 0.05141 + \cdots + 0.04914}{10} = 0.048334 \approx 0.04833s$
	$\%\delta\overline{\Delta t} = \frac{0.5(\Delta t_{max} - \Delta t_{min})}{\overline{\Delta t}} \cdot 100$ $\%\delta\overline{\Delta t} = \frac{0.5(0.05228 - 0.04518)}{0.048334} \cdot 100 = 7.34473\cdots \approx 7\%$

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Period, T	$$T = \frac{0.5(t_{blocked} + t_{unblocked})}{49.75}$$ $$T = \frac{0.5(62.48052+62.52847)}{49.75} = 1.2563717\cdots \approx 1.256372\text{s}$$
	$$\delta T = T\left(\frac{\delta t_{blocked}}{t_{blocked}} + \frac{\delta t_{unblocked}}{t_{unblocked}}\right)$$ $$\delta T = (1.2563717\cdots)\left(\frac{0.00001}{62.48052} + \frac{0.00001}{62.52847}\right) = 4.0201 \times 10^{-7} \approx 0.000001s$$
Average period, $\bar{T}$	$$\bar{T} = \frac{\sum T}{10}$$ $$\bar{T} = \frac{(1.2563717\cdots) + \cdots + (1.2562001\cdots)}{10} = 1.2561620\cdots \approx 1.256162s$$
	$$\%\delta\bar{T} = \frac{0.5(T_{max} - T_{min})}{T} \cdot 100$$ $$\%\delta\bar{T} = \frac{0.5(1.256557 - 1.255766)}{(1.2561620\cdots)} \cdot 100 = 0.0314848\cdots \approx 0.03\%$$
Dampinig constant, b	$$b = -\frac{\ln\left(\frac{2r}{\Delta t\sqrt{lg}sin\theta_0}\right)}{TN}$$ $$b = -\frac{\ln\left(\frac{2 \cdot 7.0 \cdot 10^{-3}}{0.048334 \cdot \sqrt{0.40 \cdot 9.8} \cdot \sin\left(\frac{\pi}{18}\right)}\right)}{(1.2561620\cdots) \cdot 49.75}$$ $$= 0.00274266\cdots \approx 0.0027kgm^{-1}s^{-1}$$
	$$\%\delta b = \frac{\%\delta L}{L} + \%\delta\bar{T}$$ $$where: L = \frac{2r}{\Delta t\sqrt{lg}sin\theta_0}$$ $$and: \%\delta L = \left(\frac{\delta r}{r} + \frac{\delta\Delta t}{\Delta t} + \frac{1}{2}\cdot\frac{\delta l}{l} + \frac{\lvert cos\theta_0\rvert \cdot \delta\theta_0}{\theta_0}\right)\cdot 100$$ $$\%\delta b = \frac{7.01899\cdots}{0.842485\cdots} + 0.0314848\cdots = 8.36278 \approx 8\%$$ $$where: L = \frac{2 \cdot 7.0 \cdot 10^{-3}}{0.048334 \cdot \sqrt{0.40 \cdot 9.8} \cdot \sin\left(\frac{\pi}{18}\right)} = 0.842485\cdots \approx 0.84$$ $$and: \%\delta L = \left(\frac{0.1 \cdot 10^{-3}}{7.0 \cdot 10^{-3}} + \frac{0.00002}{0.048334} + \frac{1}{2}\cdot\frac{0.5 \cdot 10^{-2}}{0.40} + \frac{\left\lvert\cos\left(\frac{\pi}{18}\right)\right\rvert \cdot \left(\frac{\pi}{360}\right)}{\left(\frac{\pi}{18}\right)}\right)\cdot 100$$ $$= 7.01899\cdots \approx 7\%$$

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Theoretical damping constant, b	$$b_T = \frac{3\pi r\eta}{m}$$ $$b_T = \frac{3\pi \cdot 7.0 \cdot 10^{-3} \cdot 1.849 \cdot 10^{-5}}{13.80 \cdot 10^{-3}} = (8.83949\cdots) \cdot 10^{-5} \approx 8.8 \cdot 10^{-5} kgm^{-1}s^{-1}$$
	$$\%\delta b_T = \left(\frac{\delta r}{r} + \frac{\delta m}{m}\right) \cdot 100$$ $$\%\delta b_T = \left(\frac{0.1 \cdot 10^{-3}}{7.0 \cdot 10^{-3}} + \frac{0.01}{13.80}\right) \cdot 100 = 1.50104 \approx 1.5\%$$
Standard deviation for Δt, $\sigma_{\Delta t}$	$$\sigma_{\Delta t} = \sqrt{\frac{\sum(\overline{\Delta t} - \Delta t)^2}{10}}$$ $$\sigma_{\Delta t} = \sqrt{\frac{(0.048334 - 0.04795)^2 + \cdots + (0.048334 - 0.04914)^2}{10}}$$ $$= 0.00248443 \approx 0.00248s$$
Standard deviation for T, σ_T	$$\sigma_T = \sqrt{\frac{\sum(\overline{T} - T)^2}{10}}$$ $$\sigma_T = \sqrt{\frac{(1.2563717\cdots - 1.256372)^2 + \cdots + (1.2563717\cdots - 1.255922)^2}{10}}$$ $$= 0.000258201 \approx 0.00026s$$
Percentage error, $\%_{error}$	$$\%_{error} = \left\lvert\frac{theoretical\ value - experimental\ value}{theoretical\ value}\right\rvert \cdot 100 = \left\lvert\frac{b_T - b}{b_T}\right\rvert \cdot 100$$ $$\%_{error} = \left\lvert\frac{(8.83949\cdots) \cdot 10^{-5} - (0.00274266\cdots)}{(8.83949\cdots) \cdot 10^{-5}}\right\rvert \cdot 100 = 3002.74\cdots \approx 3\dot{0}00\%$$

Table 8: List of Formulas, Sample Calculations, and Uncertainties

3.3 Graphing & Data Analysis

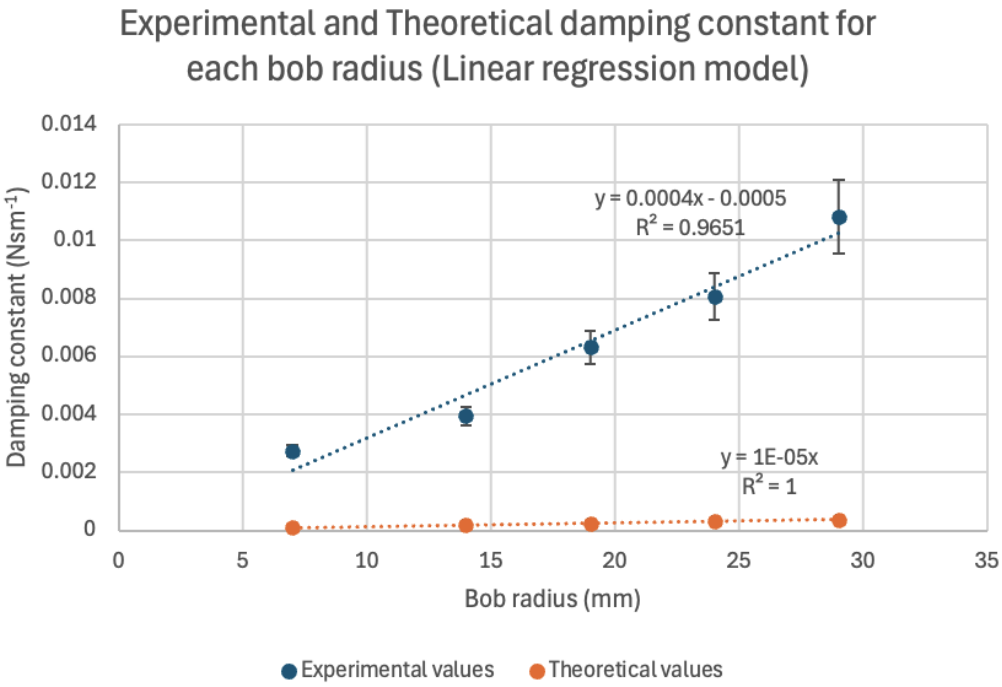


Figure 4: A scatter plot showing the experimental and theoretical relationship between bob radius and damping constant with linear regression model (Some error bars are too small to be seen)

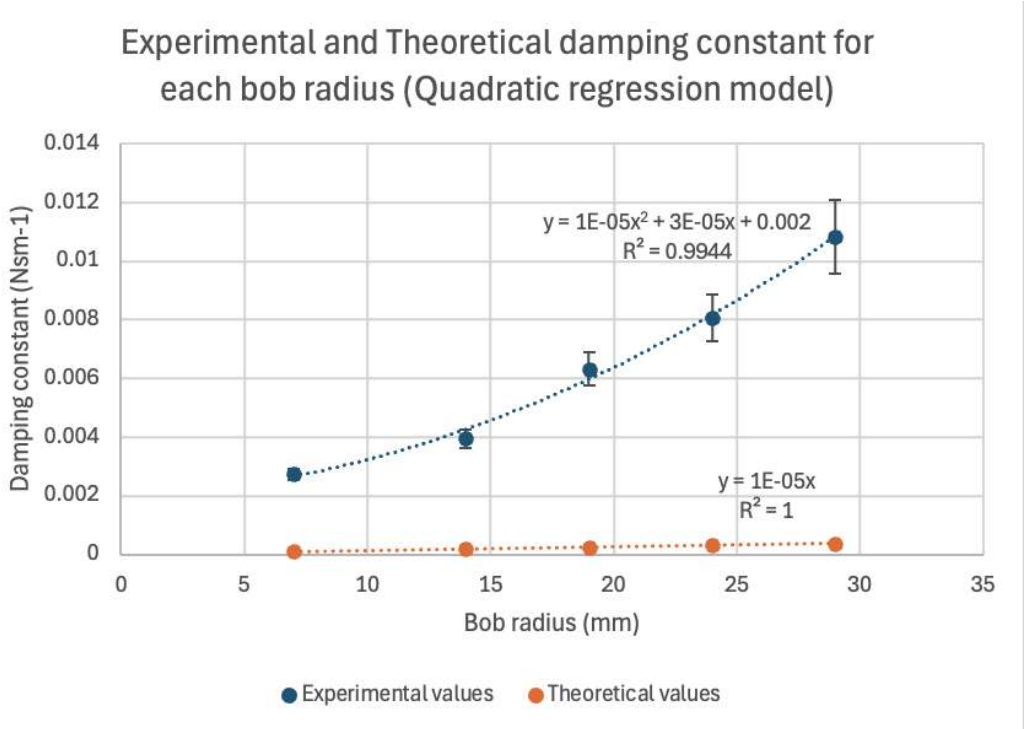


Figure 5: A scatter plot showing the experimental and theoretical relationship between bob radius and damping constant with quadratic regression model (Some error bars are too small to be seen)

Figures 4 and 5 indicate the linear and quadratic regression model for the experimental values of damping constant as well as the theoretical values estimated by the Stoke's law.

As shown in Figure 4, the experimental results exhibited a very strong linear correlation between bob radius and damping constant, as evidenced by the very high coefficient of determination (R^2) of 0.9651 for the linear regression model. However, as the line of best fit does not pass through the error bars for some of the data points, it may not be the most appropriate regression model for this dataset.

In contrast, Figure 5 adopts quadratic regression model, which demonstrates a slightly stronger correlation with R^2 value of 0.9944. Since the line of best fit passes through the error bars for all the data points, this quadratic model appears more consistent with the obtained dataset.

4 Conclusion

The aim of this experiment was to investigate the relationship between the bob radius of a simple pendulum and the damping constant. From the consideration of all the collected data, it can be concluded that there is a positive correlation between bob radius and damping constant, although the exact quantitative relationship between the two variables remains uncertain.

Both of the linear and quadratic regression model for the obtained experimental values of damping constant in Figure 4 and 5 have demonstrated a very strong positive correlation between bob radius and damping constant, supporting the aforementioned conclusion.

However, while the Stoke's law predicts a linear proportionality between bob radius and damping constant, the comparison of linear and quadratic regression models indicated a greater suitability of the quadratic model for the obtained dataset from this experiment, refuting the proposed hypothesis.

It should also be noted that the experimental values of damping constant deviated substantially from the theoretical values estimated by the Stoke's law, as shown by the percentage errors ranging around 2000-3000%. This implies the presence of other forces acting against the restoring force, such as the friction between the pivot and the string as well as the drag force on the string. In fact, it is reported that the drag force on the string can play more significant role than the drag force on the bob in damping effect depending on the conditions**Invalid source specified.**. This contradicts one of the key assumptions of this experiment that the damping effect is predominantly due to the drag force acting on the bob, undermining the validity of the derived conclusion.

The suitability of quadratic model over linear model and the extremely high percentage error also suggest that the application of Stoke's law for modelling the drag force on the bob as well as the use of Eq. (13) for estimating b might have not been appropriate. This will be discussed further in the evaluation section.

The experimental results also exhibited a high degree of random errors. This can be inferred from the standard deviations $\sigma_{\Delta t}$ and σ_T for time in gate Δt and period T. While the measurement uncertainties for Δt and T are ±0.00002 and ±0.00001, respectively, the averages of standard deviations are $\sigma_{\Delta t} = 0.00047$ and $\sigma_T = 0.00792$ which are about 20-800 times larger than the instrumental uncertainties, making it difficult to reliably determine the exact quantitative relationship.

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5 Evaluation

1. 3D motion of bob

   While it was assumed that the bob only moves on a vertical 2D plane, in the experiment, the bob demonstrated a 3D motion resembling an ellipse for some of the trials. This implies that some of the mechanical energy existed as rotational movement on a horizontal plane in some trials, causing random errors.

   Since it is extremely difficult to prevent the rotational movement of the bob in the horizontal direction, this random error may be mitigated by excluding the data obtained from trials in which the bob demonstrated an elliptical motion.

2. Drag force on string and friction

   While this experiment assumes that the damping effect is predominantly due to the drag force on the bob, other factors may have also contributed to the damping effect, causing systematic errors. One of them is the drag force on the string, as it is reported that it can play a more significant role than the drag force on the bob under some conditions**Invalid source specified.**. Another factor may be the friction between the pivot and the string, which causes energy loss.

   The effect of drag force on string can be mitigated by using a thinner string. Similarly, the effect of friction can be reduced by tightly tying the string to the pivot so that the string does not move, and thus there is no work done against friction. It is also possible to reduce the relative effect of the other damping sources by increasing the bob radius so that the damping effect due to drag on the bob becomes relatively significant.

3. The validity of application of Stoke's law

   One of the key assumptions was that bob's velocity is low enough that the airflow around the bob is laminar (non-turbulent) and the effect of pressure drag can be assumed negligible. This assumption ensures that the Stoke's law is applicable, as it only applies to a laminar flow where the Reynolds number is below 1 (Dey, Ali, & Padhi, 2019). Reynolds number is a quantitative measure of flow characteristics to distinguish between laminar and turbulent region**Invalid source specified.**, given by

$$Re = \frac{\rho v L}{\mu} \tag{18}$$

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where ρ is fluid density (kgm^{-3}), v is relative flow velocity (ms^{-1}), L is characteristic length (m), and μ is dynamic viscosity of fluid (Nsm^{-2}). By substituting appropriate values**Invalid source specified.** of $\rho = 1.184 kgm^{-3}$ at 298.15K, $v = v_{max} = \frac{2r}{\Delta t} = \frac{2 \cdot 7.0 \cdot 10^{-3}}{0.04833}$, $L = 2r = 2 \cdot 7.0 \cdot 10^{-3}$, and $\mu = 1.849 \cdot 10^{-5} Nsm^{-2}$, we can roughly estimate the Reynolds number of the airflow around the bob to be

$$Re = \frac{1.184 \cdot \left(\frac{2 \cdot 7.0 \cdot 10^{-3}}{0.04833}\right) \cdot 2 \cdot 7.0 \cdot 10^{-3}}{1.849 \cdot 10^{-5}} = 259.689 \cdots \approx 260 \quad (19)$$

Since Reynolds number of 260 is way beyond the condition of $Re < 1$, this suggests that the Stoke's law is no longer accurate under these experimental conditions. Since the damping model assumed valid application of Stoke's law, the experimental values of damping constants calculated using the model may not accurately represent the true damping of the pendulum.

Alternatively, we may apply the drag equation, which is applicable to flow of high Reynolds number where the pressure drag is dominant over viscous drag. The drag equation **Invalid source specified.** is given by

$$F_{drag} = \frac{1}{2} \rho v^2 c_d A \quad (20)$$

where ρ is fluid density (kgm^{-3}), v is relative flow velocity (ms^{-1}), c_d is drag constant (dimensionless), and A is reference area (m^2). Since the reference area A is given by $A = \pi r^2$, the drag equation becomes

$$F_{drag} = \frac{1}{2} \rho v^2 c_d \pi r^2 \quad (21)$$

Since $F_{drag} = \gamma v$ and $b = \frac{\gamma}{2m}$, using the drag equation, we can derive the relationship between the bob radius r and drag constant b as

$$b = \frac{\rho v c_d \pi r^2}{2m} \quad (22)$$

This expression suggests the relationship $b \propto r^2$, which aligns with the experimental results that favors quadratic relationship between b and r. However, it should be noted that the equations $F_{drag} = \gamma v$ and $b = \frac{\gamma}{2m}$ used to derive Eq. (22) were built on the assumption that $F_{drag} \propto v$. Since the drag equation suggests that $F_{drag} \propto v^2$, it contradicts the underlying assumption. Therefore, it is not appropriate to simply replace the Stoke's law with drag equation, but rather a new model for the motion of damped simple pendulum need to be constructed by modelling drag force on bob as $F_{drag} = -\gamma v^2$.

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6 Appendix

6.1 Obtaining a General Solution to the Differential Equation in Eq. (9)

Recall that, as in Eq. (9), the motion of a dumped simple pendulum can be expressed by

$$m\frac{d^2x}{dt^2}+\gamma\frac{dx}{dt}+\frac{mg}{l}x=0 \tag{23}$$

By diving the both sides of Eq. (23) by m, we obtain

$$\frac{d^2x}{dt^2}+\left(\frac{\gamma}{m}\right)\frac{dx}{dt}+\frac{g}{l}x=0 \tag{24}$$

For an ideal simple pendulum, it is known that period T is given in terms of string length l and gravitational acceleration g as

$$T=2\pi\sqrt{\frac{l}{g}} \tag{25}$$

As mentioned in the Appendix 6.3, natural angular frequency ω_0 and damped angular frequency ω' are approximately equal. Therefore, damped angular frequency can be written as

$$\omega'=\omega_0=\frac{2\pi}{T}=\sqrt{\frac{g}{l}} \tag{26}$$

Thus, by substituting Eq. (26) into the Eq. (24), we obtain

$$\frac{d^2x}{dt^2}+\left(\frac{\gamma}{m}\right)\frac{dx}{dt}+\omega_0^2x=0 \tag{27}$$

This now becomes a second-order linear homogeneous differential equation. We can assume that that the solution to this differential equation is in the form

$$x(t)=Qe^{\lambda t} \tag{28}$$

where Q and λ are arbitrary constants**Invalid source specified.**.

Therefore, the 1st and 2nd derivatives can be written as

$$\frac{dx}{dt}=\lambda Qe^{\lambda t} \tag{29}$$

$$\frac{d^2x}{dt^2}=\lambda^2Qe^{\lambda t} \tag{30}$$

Now, substituting Eq. (28), (29), and (30) back into Eq. (27) and rearranging it yields

$$\lambda^2Qe^{\lambda t}+\left(\frac{\gamma}{m}\right)\lambda Qe^{\lambda t}+\omega_0^2Qe^{\lambda t}=0 \tag{31}$$

$$Qe^{\lambda t}\left\{\lambda^2+\left(\frac{\gamma}{m}\right)\lambda+\omega_0^2\right\}=0 \tag{32}$$

Since $Qe^{\lambda t}\neq 0$ for $t\in\mathbb{R}$, $Qe^{\lambda t}$ term can be factored out, which gives

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$$\lambda^2 + \left(\frac{\gamma}{m}\right)\lambda + \omega_0^2 = 0 \tag{33}$$

This Eq. (33) can be seen as a quadratic equation in terms of λ. Using the quadratic formula, the roots of this quadratic equation can be expressed as

$$\lambda = \frac{-\left(\frac{\gamma}{m}\right) \pm \sqrt{\left(\frac{\gamma}{m}\right)^2 - 4\omega_0^2}}{2} = -\frac{\gamma}{2m} \pm \frac{\sqrt{\frac{\gamma^2}{m^2} - 4\omega_0^2}}{2} = -\frac{\gamma}{2m} \pm \sqrt{\frac{\gamma^2}{4m^2} - \omega_0^2} \tag{34}$$

For an underdamped pendulum, we consider the case where $\frac{\gamma^2}{4m^2} - \omega_0^2 < 0$ (Chasnov, n. d.). To make the term inside the square root positive, λ can be expressed using an imaginary number i as

$$\lambda = -\frac{\gamma}{2m} \pm i\sqrt{\omega_0^2 - \frac{\gamma^2}{4m^2}} \tag{35}$$

Now we define damped angular frequency ω', where $\omega' = \sqrt{\omega_0^2 - \frac{\gamma^2}{4m^2}}$, giving

$$\lambda = -\frac{\gamma}{2m} \pm i\omega' \tag{36}$$

By substituting the two roots back into Eq. (28), the general solution $x(t)$ can be written as the sum of the two solutions $x_1(t)$ and $x_2(x)$, each multiplied by arbitrary constants C_1 and C_2, respectively (Dawkins, 2023), which gives

$$x(t) = C_1 e^{\left(-\frac{\gamma}{2m} + i\omega'\right)t} + C_2 e^{\left(-\frac{\gamma}{2m} - i\omega'\right)t} = e^{-\frac{\gamma}{2m}t}\left(C_1 e^{i\omega' t} + C_2 e^{-i\omega' t}\right) \tag{37}$$

Using the Euler's formula $e^{i\theta} = cos\theta + isin\theta$, we can rewrite Eq. (37) as

$$x(t) = e^{-\frac{\gamma}{2m}t}\{C_1 \cos(\omega' t) + C_1 isin(\omega' t) + C_2 \cos(-\omega' t) + C_2 isin(-\omega' t)\} \tag{38}$$

$$x(t) = e^{-\frac{\gamma}{2m}t}\{C_1 \cos(\omega' t) + C_1 isin(\omega' t) + C_2 \cos(\omega' t) - C_2 isin(\omega' t)\} \tag{39}$$

$$x(t) = e^{-\frac{\gamma}{2m}t}\{(C_1 + C_2)cos(\omega' t) + (C_1 - C_2)isin(\omega' t)\} \tag{40}$$

Now we introduce two arbitrary constants A and B where $A = (C_1 + C_2)$, and $B = (C_1 - C_2)i$. This simplifies Eq. (40) into

$$x(t) = e^{-\frac{\gamma}{2m}t}\{Acos(\omega' t) + Bsin(\omega' t)\} \tag{41}$$

Recall that a general solution $x(t)$ should yield a real number, as it describes the displacement of a bob. Therefore, constants $C_1$and C_2 must be a conjugate pair, so that both constant A and B are real. This is proven by letting $C_1 = \alpha + \beta i, C_2 = \alpha - \beta i$ where $\alpha, \beta \in \mathbb{R}$, which gives expressions for A and B as

$$A = (\alpha + \beta i) + (\alpha - \beta i) = 2\alpha \tag{42}$$

$$B = \{(\alpha + \beta i) - (\alpha - \beta i)\}i = 2\beta i^2 = -2\beta \tag{43}$$

Hence, A and B are both real numbers, and thus the general solution $x(t)$ in Eq. (41) is now entirely expressed in terms of real numbers.

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Now let $A = x_0 cos\phi$ and $B = x_0 sin\phi$. By substituting this into Eq. (41) and using trigonometric identity, we can obtain the final form of general formula $x(t)$ as

$$x(t) = e^{-\frac{\gamma}{2m}t}\{x_0 cos\phi\ cos(\omega' t) + x_0 sin\phi\ sin(\omega' t)\} \tag{44}$$

$$x(t) = x_0 e^{-\frac{\gamma}{2m}t} \cos(\omega' t - \phi) \tag{45}$$

This Eq. (45) describes the motion of a damped simple pendulum.

6.2 Deriving an Expression for b with Measurable Variables

To derive an expression for damping constant b with measurable variables in the experiment, we will start by considering the energy of a damped simple pendulum.

The total energy of a damped simple pendulum is given by

$$E_{total} = \frac{1}{2} m\omega'^2 x_{max}^2 \tag{46}$$

Since the kinetic energy at the equilibrium position where the bob achieves its maximum velocity v_{max} equals the total mechanical energy, we obtain

$$\frac{1}{2} m v_{max}^2 = \frac{1}{2} m\omega'^2 x_{max}^2 \tag{47}$$

By eliminating common factors and rearranging, we obtain an expression for amplitude $x_{max}\ as$

$$x_{max} = \frac{v_{max}}{\omega'} \tag{48}$$

As in Eq. (26), the damped angular frequency of a simple pendulum ω' can be approximated by

$$\omega' = \sqrt{\frac{g}{l}} \tag{49}$$

By substituting Eq. (49) into Eq. (48), we obtain

$$x_{max} = v_{max}\sqrt{\frac{l}{g}} \tag{50}$$

By equating Eq. (12) and (50), and rearranging, we can obtain the equation for damping constant b in terms of velocity at equilibrium position v_{max} at a given time t as follows

$$v_{max}\sqrt{\frac{l}{g}} = x_0 e^{-bt} \tag{51}$$

$$e^{-bt} = \frac{v_{max}}{x_0}\sqrt{\frac{l}{g}} \tag{52}$$

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$$-bt = \ln\left(\frac{v_{max}}{x_0}\sqrt{\frac{l}{g}}\right) \tag{53}$$

$$b = -\frac{\ln\left(\frac{v_{max}}{x_0}\sqrt{\frac{l}{g}}\right)}{t} \tag{54}$$

For a small initial angular displacement θ_0, using trigonometry, the initial horizontal displacement x_0 can be approximated by

$$x_0 = lsin\theta_0 \tag{55}$$

Since the velocity of the bob is maximum at the equilibrium position, assuming that the velocity at around the equilibrium position is apporximately constant over the short time interval Δt, the maximum velocity v_{max} of the bob can be estimated by the time taken for the bob to move past the equilibrium position Δt with the following expression:

$$v_{max} = \frac{2r}{\Delta t} \tag{56}$$

where r is the bob radius (m).

By writing expressions for angular displacement θ, time t can be expressed as

$$\theta = \omega' t = 2\pi N \tag{57}$$

$$t = \frac{2\pi}{\omega'}N = TN \tag{58}$$

where N is number of completed oscillations and T is the period of a pendulum.

Substituting Eq. (55), (56) and (58) into Eq. (54) yields an final form of an expression for b as

$$b = -\frac{\ln\left(\frac{2r}{\Delta t\ l\ sin\theta_0}\sqrt{\frac{l}{g}}\right)}{TN} = -\frac{\ln\left(\frac{2r}{\Delta t\sqrt{lg}sin\theta_0}\right)}{TN} \tag{59}$$

6.3 Difference in Natural and Damped Angular Frequency

As mentioned in the previous section, damped angular frequency ω' is given by

$$\omega' = \sqrt{\omega_0^2 - \frac{\gamma^2}{4m^2}} \tag{60}$$

where ω_0 is the natural angular frequency, $\gamma = 6\pi r\eta$, and m is the mass of bob.

Using this equation, we are able to calculate the theoretical difference between natural and damped angular frequency in this experiment. Out of the several variables involved in the equation, bob radius r is varied among different trials. Since it can be inferred from the equation that the difference between the

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two angular frequencies is the greatest when r is maximized. Therefore, the largest bob radius, $29mm$, is used in this estimation. The dynamic fluid viscosity of air is assumed $1.849 \cdot 10^5\ kgm^{-1}s^{-1}$ at 298.15K. Thus, letting $\omega_0 = 1\ rads^{-1}$ and substituting $r = 0.029m, m = 0.01380kg, \eta = 1.849 \cdot 10^5 kgm^{-1}s^{-1}$ gives

$$\omega' = \sqrt{1^2 - \frac{(6\pi \cdot 0.029 \cdot 1.849 \cdot 10^{-5})^2}{4 \cdot 0.01380^2}} = 0.9999999329 \cdots \approx 1\ rads^{-1} = \omega_0 \qquad (60)$$

This shows that damped angular frequency ω' is approximately equal to natural angular frequency ω_0 under these experimental conditions.

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